

## Course Outline

## CSI141 Programming Principles (3 credits)

## Course Staff

|                     |                                                                                       |
|---------------------|---------------------------------------------------------------------------------------|
| Lecturer            | <b>Tallman Nkgau</b><br>Office: 247-285<br>Phone: 355 2260<br>Email: nkgautz@ub.ac.bw |
| Teaching Assistants | <b>TBA</b>                                                                            |

## Teaching and Learning

**Lectures (in-person):** Mondays: 1500-1650 233-G12  
Thursdays: 1300-1450 230-G5

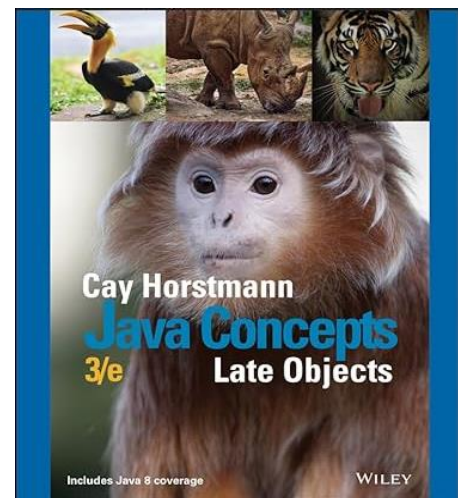
**Lab:** Tuesday 0900 – 1200, 1200 – 1500, 1500 – 1800, 247-293/294/295/296/297 (you choose 1 session)

**Office Hours:** Tuesdays 1000 – 1200 or by appointment.

**Course website:** [www.cs.ub.bw/moodle](http://www.cs.ub.bw/moodle)

## Course resources

- **Java Concepts, Late Objects, 3<sup>rd</sup> Edition, Cay Horstmann (required textbook)**



- Introduction to Programming with Java  
<https://runestone.academy/ns/books/published/csjava/index.html>
- Introduction to Programming Using Java, Version 9.0, JavaFX Edition, David J. Eck  
<https://math.hws.edu/javanotes/>
- The Java Tutorials, Oracle <https://docs.oracle.com/javase/tutorial/>
- Think Java: How to Think Like a Computer Scientist 2e, Allen Downey and Chris Mayfield, O'REILLY <https://greenteapress.com/wp/think-java-2e/>
- The Java Visualizer [https://cscircles.cemc.uwaterloo.ca/java\\_visualize/#](https://cscircles.cemc.uwaterloo.ca/java_visualize/#)

You are responsible for material covered in the lectures, laboratories, and any assigned reading. The laboratories are essentially programming/problem solving sessions and **attendance is mandatory**.

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## Course Website

<http://www.cs.ub.bw/moodle/course/view.php?id=157>

## Course Evaluation

Please note the following Faculty of Science Special Regulation (Page 176 of the 2022/2023 Undergraduate Academic Calendar).

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### 23.9 Minimum CA to Qualify for Examinations

**23.9.1 A minimum continuous assessment mark of 40% is required to sit for final examinations.**

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|----------------------------------|-----|----------------------------------------------------------|
| <b>3-Hours Final Examination</b> | 50% | UB Scheduled Exam                                        |
| <b>2-Hours Midterm Test</b>      | 15% | Either the week before short vac or week after short vac |
| <b>Graded Labs</b>               | 20% |                                                          |
| <b>Lab Tests x 2</b>             | 10% |                                                          |
| <b>Participation</b>             | 5%  | Usually based on performance and attendance in the labs  |

## Course Synopsis

Overview of problem solving; the programming process; high level languages; data types, input/output, control structures; functions and classes; libraries; files and streams; simple data structures (arrays and ArrayLists); practical problem-solving exercises.

## Pre-requisite

Curiosity!

## Aim

Computers and computing gadgets are ubiquitous. All aspects of our lives depend on computing technology. Learning how to use technology and innovating new ways of using it is imperative. Programming is now regarded as a fundamental skill as it allows us to effectively use computing technology and create new solutions. This course aims to introduce students to the basic concepts and principles of programming using a high-level language. Its primary intention is to develop key programming and problem solving skills. It is the first course in the sequence of courses that aim to impart software development skills.

## Learning Outcomes

At the end of this course, students should be able to:

- Use the Java compiler, a code editor and hand debug programs.

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- b) Develop algorithms to solve basic computational problems.
- c) Write a program for a given computational problem.
- d) Implement, debug and test programs.

## Tentative Schedule (May and will change)

| Week    | Topics                                                                                                                                                                                                                                                                     | Reading                        | Exercises                                                                                                            |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 1       | Course info<br>Course learning outcomes<br>Course assessment criteria<br>Course rules<br>Why program?<br>Anatomy of a computer<br>Data representation                                                                                                                      | Chapter 1.1 – 1.4              | R1.4<br>Quiz 1                                                                                                       |
| 2-3     | Programming process<br>Programming environment<br>Everyday data<br>Java primitive data types<br>Variables, & assignment,<br>keyboard/console input/output<br>String Type, variables, & assignment,<br>keyboard/console input/output<br>1D & 2D arrays of Numbers & Strings | Chapter 1.5 – 1.6<br>Chapter 2 | Quiz 2<br>E1.3, E1.3, E1.4, E1.5<br>E1.11, E1.20<br>R2.1 – R2.8<br>R2.11, R2.15, R2.22<br>E2.4,<br>P2.1, P2.3, P2.14 |
| 4-6     | Boolean expression<br>Comparison operators<br>1D arrays of Booleans<br>Selection statements                                                                                                                                                                                | Chapter 3                      | Quiz 3<br>R3.1 – R3.8<br>R3.30 – R3.32<br>E3.2, E3.5, E3.7<br>P3.1, P3.4, P3.7, P3.17                                |
| 7 - 9   | Repetition: while loop, for loop<br>File input/output<br>1D, 2D arrays and ArrayLists                                                                                                                                                                                      | Chapter 4<br>Chapter 6         | Quiz 4<br>R4.1 – R4.7, R4.13<br>R4.16 – R4.17<br>P4.1 – P4.3<br>R6.1 – R6.4<br>E6.1, E6.2<br>P6.1, P6.2              |
| 10 - 13 | Functions & Libraries<br>User-defined data types – Java classes                                                                                                                                                                                                            | Chapter 5<br>Chapter 8         | Quiz 5<br>R5.2, R5.4, R5.5, R5.9<br>E5.2, E5.3<br>P5.14<br>E8.7, E8.9<br>P8.6, P8.7                                  |

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### Laboratory submissions

Assessed labs are submitted during the lab. The assessed lab submission should be done 10 minutes before the lab session ends. In case of technical problems, **inform the teaching assistants in charge as soon as possible and before the end of the lab session**. Submissions are only allowed through the provided submission links on the course website. No other method of submission is allowed without the prior permission of the course staff.

### Handouts

All **handouts** will be made available on the course website. You are responsible for printing your own hardcopies of the handouts. Please check the course website regularly as course announcements will also be put there. **Failure to notice an announcement will not be accepted as an excuse for missing assessments.**

### Collaboration Policy

You are encouraged to form study groups as this often leads to mastery of the material. However,

You **MUST** write up or implement each problem solution yourself without assistance.

Laboratories are to be done individually except when peer assistance is allowed. **All assessments are done individually.** You must always acknowledge your sources if you did not come up with the solution yourself.

**Plagiarism** will not be tolerated in the course and will be dealt with according to university rules and regulations. The academic honesty policy for students' document is available online at our course website and should be read and understood.

We will be using the university provided software for checking similarity between submitted solutions. If the similarity is too high between solutions from different students, the course staff will verify that, and students may also be asked to explain their submissions.

### Communication & Assistance

The primary mode for communicating with the course lecturer is by email – [nkgautz@ub.ac.bw](mailto:nkgautz@ub.ac.bw). If using email, students must use their **university assigned email account**. We will ONLY respond if the email used is from your university assigned email account. Please **DO NOT** use Moodle to email teaching staff.

Students who require assistance should see the course lecturer during office hours. Contact information for Teaching Assistants will be made available once they are on ground.